Final Design Report

EG-499 Engineering Capstone II

TEAM THERMOCLINE

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Abstract:

SNHU's final capstone course required students to collaborate across engineering disciplines to design, fabricate, test, analyze, optimize, and present a working product that satisfied customer requirements. Each student brought their unique mechanical and electrical experience to help develop the final product: a thermally-controlled PCB testing chamber that delivered efficient and highly successful performance. Team Thermocline was originally presented with several driving requirements provided by the customer – an electrical engineering professor, which described product dimension, function, and performance criteria. From this, the group developed a conceptual design, including materials, components, control features, and the expected assembly integration process. Trade studies were then conducted across the project, supported by external consultations with engineering, HVAC, and manufacturing/machining professionals to approve – or improve – the product design. Initial studies included prototype electronic boards and mechanical parts to test supply chain capabilities and electronic performance feasibility. The team built on these initial steps by designing the mechanical structure, creating the main electronic control board, sourcing thermal-fluid components, ordering critical pieces, and 3D-printing/laser-cutting parts. Assembly was supplemented with iterative designing, prototyping, and functional testing of individual subsystems. The end of the course brought performance testing to observe alignment with customer requirements, and presenting to relevant project stakeholders. Proper documentation was completed throughout the project, showing clear product development history. By the end, all system and driving customer requirements were met, achieved by persistent, insightful, and collaborative teamwork. However, one derived requirement was not met (CON-05), describing the prohibition of thermal control when the user is handling the PCB. This resulted from a lack of testing with physical PCBs inside the chamber, and implementing the necessary sensors/measurements/controls to enforce this requirement. In spite of this, the team successfully designed, built, and tested the thermally-controlled chamber that met/exceeded most requirements, and served as an enlightening experience for developing well-rounded engineers.

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1 Introduction

Figure 1; Internal Chamber Cooling Process

1.1 Mission Purpose

Design and develop a thermally controlled chamber capable of reliably evaluating electronic component performance under varying temperature conditions. The system will provide a stable, insulated environment that supports a 12"×12" printed circuit board, enables external electrical interfacing, and maintains precise temperature control within the commercial operating range of 0°C to 70°C. By incorporating accurate temperature monitoring and data acquisition, the chamber will ensure consistent, repeatable testing to meet project sponsor requirements.

1.2 Background

Thermally controlled chambers are industrial machines that measure, track, display, modify, and control environmental conditions of an enclosed system. Inside the system, products are exposed to specific environmental temperatures, and their performance and integrity is analyzed. By replicating expected or predicted thermal conditions in a relatively cheap, efficient, and simple manner, these systems advance research, assist the engineering design process, and streamline functional testing of products. Often they are equipped with heating and cooling subsystems (sometimes an intertwined mechanical process) to reach a wide range of extreme temperatures. The fundamental features of these chambers require manual user interaction for temperature setpoints and adjusting system conditions, while they also incorporating automatic programmed commands to execute highly accurate and responsive control algorithms. Typically, the entire system consists of an internal structural enclosure – for housing the tested components,

an external structural enclosure – for housing all subsystems, an electronic control system, and any other materials and devices used to achieve temperature modification. In industrial applications, these thermally-controlled chamber systems are highly reputable, known for their versatility in meeting various customer requirements for a project.

1.3 Concept of Operations

The Concept of Operations – hereafter referred to as "ConOps", outlines the intended function, environment, and operational behavior of the thermally controlled PCB testing chamber. The system is a tabletop, insulated chamber designed to accommodate 12"×12" PCBs and regulate internal temperature within the commercial range of 0–70°C while maintaining ±3°C accuracy. It operates in an indoor laboratory environment with stable power, proper ventilation, and access to PC-based data logging. The chamber supports multiple operating modes—including Shutdown, Standby, Start, and Safe states—and allows users to insert test boards, connect measurement cables, and monitor temperature performance locally or remotely. The ConOps identifies primary stakeholders such as engineers, maintenance personnel, faculty, and material suppliers, all of whom contribute to system design, operation, and lifecycle support. It also evaluates operational impacts, including thermal hazards, electrical safety concerns, ventilation requirements, spatial constraints, and data management needs. Finally, the ConOps provides a structured risk assessment addressing potential failures in design, fabrication, operation, and maintenance, assigning likelihood and consequence values to prioritize mitigation efforts. Collectively, the ConOps establishes a clear operational framework that informs system requirements, guides engineering decisions, and ensures safe, reliable, and traceable thermal testing of electronic components.

2 System Requirements

Requirement Type	Number	Requirement	Verification Method
Design	1.1	The system shall accommodate a 12x12" PCB as the device to be tested.	Demonstration
Design	1.2	The system shall include insulation designed for external safety according to ASTM C1055.	Inspection
Design	1.3	The system's design shall be safe for continual operation according to NFPA-86.	Inspection
System	2.1	The system shall include a port to allow electrical cables to enter and leave for stimulus and measurement of the PCB.	Inspection
System	2.2	The system shall comply with electrical grounding requirements.	Inspection
System	2.3	The system shall be able to operate on a standard wall-outlet power of 120V (AC).	Demonstration
System	2.4	The system shall comply with appropriate NEC and UL rules governing powered and plug-in devices.	Inspection
Functional	3.1	The system shall monitor temperature inside the chamber.	Demonstration

Functional	3.2	The system shall control temperature inside the chamber.	Demonstration
Functional	3.3	The system shall remain within a specified humidity range (0% - 90%).	Demonstration
Functional	3.4	The system shall provide data output.	Demonstration
Interface	4.1	The system shall include an HMI for the control system.	Inspection
Interface	4.2	The system shall include separate operating/control modes.	Demonstration

Table 1; System Requirements

3 Chosen Design

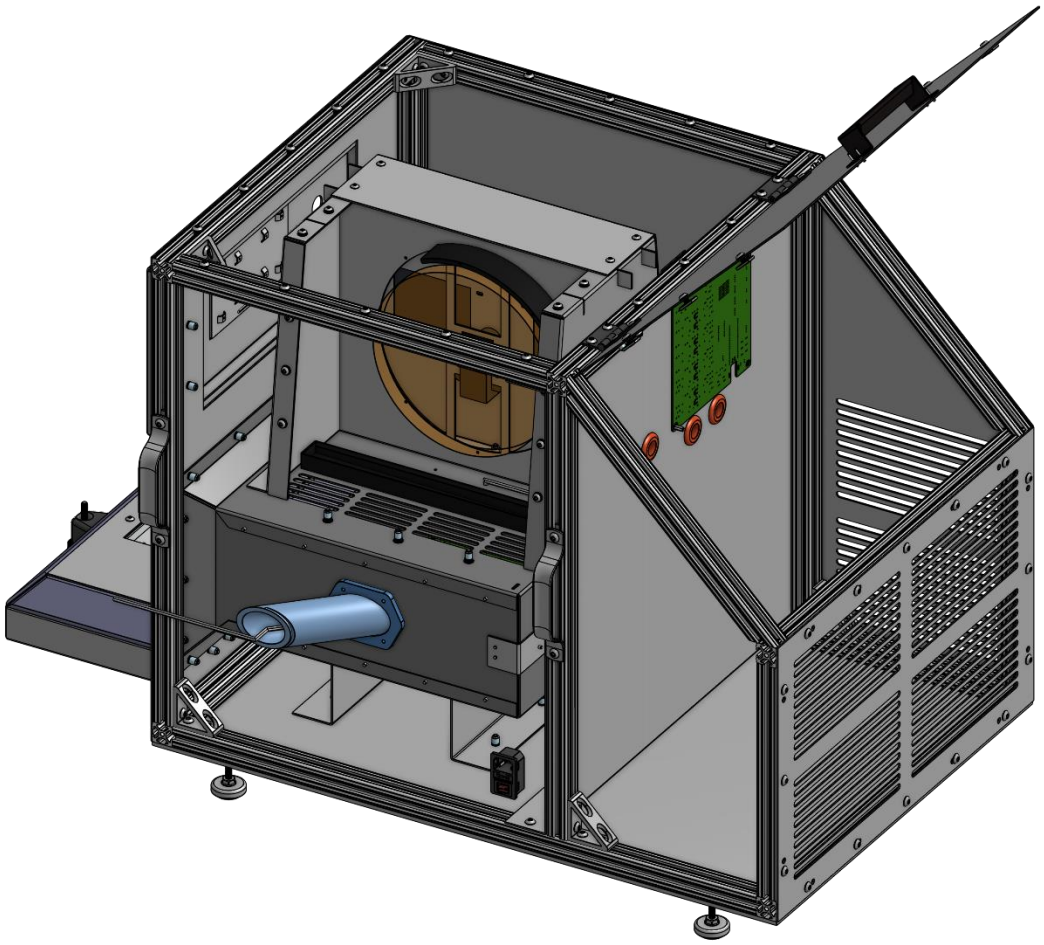

Figure 2; CAD System Overview

The final design is a fully integrated temperature-controlled chamber developed to evaluate printed circuit boards under controlled environmental conditions. The system is designed to accommodate a 12"×12" PCB while maintaining precise control of internal conditions across a temperature range of 0°C to 70°C. The chamber operates on standard 120V power and provides a stable, insulated environment for reliable and repeatable testing, including real-time monitoring and data logging capabilities.

The system architecture is divided into five primary subsystems: structural, thermo-fluid, safety, electrical components, and control/software. The structural subsystem provides the physical enclosure and supports all internal components, while the thermo-fluid subsystem regulates internal temperature through integrated heating, cooling, and airflow management. The electrical subsystem supplies power and enables sensing and

data acquisition, and the control/software subsystem manages temperature regulation, operating modes, and user interaction through an HMI. The safety subsystem ensures compliance with applicable standards and protects both the user and the system during operation.

These subsystems are designed to function as a cohesive unit, with mechanical, electrical, and control elements closely integrated. The structural design supports thermal efficiency and component placement, while the thermo-fluid system depends on electrical components and control logic to maintain precise environmental conditions. The control system continuously monitors system performance and adjusts operation to maintain stability, while also enforcing safety interlocks and operational limits.

A key design decision in the system was the selection of insulation material. Denim insulation was chosen over fiberglass insulation to improve safety during assembly and handling. While fiberglass offers slightly higher thermal performance, it introduces potential hazards such as skin irritation and airborne fibers. Denim insulation provides a safer alternative with comparable insulating capability, making it a suitable choice for this application without significantly compromising thermal efficiency.

Overall, the chosen design meets the required performance, safety, and operational objectives while maintaining a practical and manufacturable system. Further details on each subsystem and how specific requirements are satisfied are provided in the following section.

4 Subsystems

This section describes each subsystem in detail, including design implementation, requirement satisfaction, and supporting trade studies used to justify key engineering decisions. Trade studies were conducted to evaluate design alternatives based on cost, performance, safety, and system integration.

4.1 Structural Subsystem

Purpose and Design Description

The structural subsystem provides the physical enclosure and support for all internal components. The chamber was designed with external dimensions not exceeding 30"×30" and internal dimensions of at least 12"×12" to accommodate the PCB and ensure adequate airflow. The enclosure includes sealed openings and joints to minimize thermal losses and maintain internal environmental control. Structural components also support the integration of insulation, thermal hardware, and electrical components.

Requirements Satisfaction

The structural subsystem satisfies the following requirements:

- STR-01 and STR-02 through controlled external and internal dimensions
- STR-03 through sealed chamber openings
- STR-04 through proper sealing of internal thermal interfaces

Trade Study – Structural Material Selection

A trade study was conducted to determine the optimal material for the enclosure and supporting structure. Materials considered included carbon steel, titanium alloys, composites, plastics, and aluminum.

The team selected **5052 H32 aluminum (0.063")** due to its:

- Low weight, improving manufacturability and handling
- Sufficient strength for structural support
- Compatibility with thermal conditions
- Cost-effectiveness and availability

Additionally, outsourcing fabrication to a manufacturer such as Send-Cut-Send improved precision, reduced lead time, and simplified manufacturing.

4.2 Thermo-Fluid Subsystem

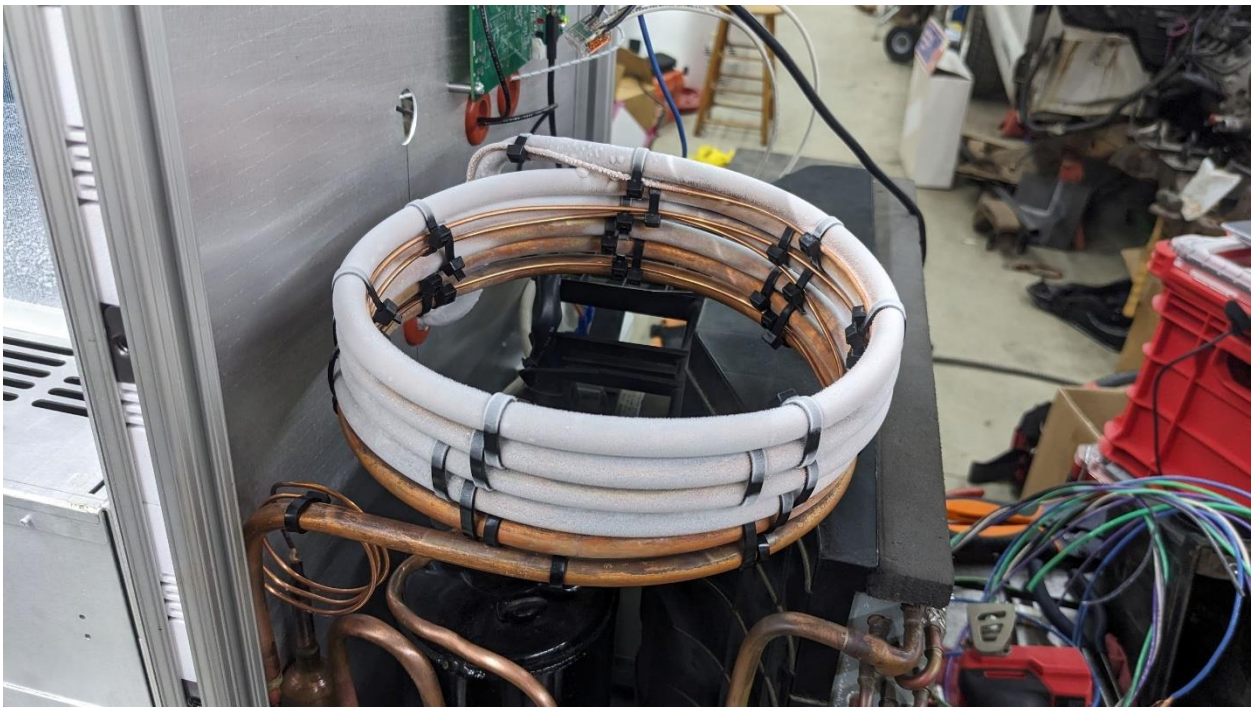

Figure 3; Copper Wiring During Cooling Procedure

Purpose and Design Description

The thermo-fluid subsystem is responsible for controlling the internal temperature of the chamber across a range of 0°C to 70°C. The system incorporates enclosed heating and cooling elements along with airflow management to ensure uniform temperature distribution across the PCB. The system is capable of sweeping the full temperature range within a 24-hour cycle.

Requirements Satisfaction

The thermo-fluid subsystem satisfies:

- THE-01 through achieving the required temperature range
- THE-02 through controlled temperature ramping
- THE-03 and THE-04 by enclosing all heating and cooling components

Trade Study – Heating Method Selection

Option	Description	Cost	Risk
Resistive Heating	Electrical resistance generates heat	~$100	Overheating/fire hazard
Inductive Heating	Magnetic field induces internal heating	~$800	EMI, complexity
Convection Heating	Heated fluid/air circulation	~$600	Non-uniform heating
Reversible Heating	Heat recovery system	~$500	Increased complexity, failure points

Table 2; Trade Study; Heating Method

Decision:
Resistive heating was selected due to its low cost, simplicity, and reliability. Although it presents overheating risks, these are mitigated through distributed temperature sensing and control logic.

Reversible heating was not selected due to increased system complexity, cost, and integration challenges.

Trade Study – Cooling Method Selection

Option	Description	Cost	Risk	Integration
R410A Refrigerant	Common AC refrigerant	~$60/lb	Moderate	Moderate
Glycol System	Liquid cooling loop	~$5000	Low	Easy
Propane	Refrigerant gas	Low	High (flammable)	Moderate
Peltier (TEC)	Semiconductor cooling	Very high	Moderate	Difficult

Table 3; Trade Study; Cooling Method

Decision:
R410A was selected due to its balance of cost, effectiveness, and availability. It was sourced from existing systems (e.g., portable AC units), making it a practical solution within budget constraints.

4.3 Electrical Components Subsystem

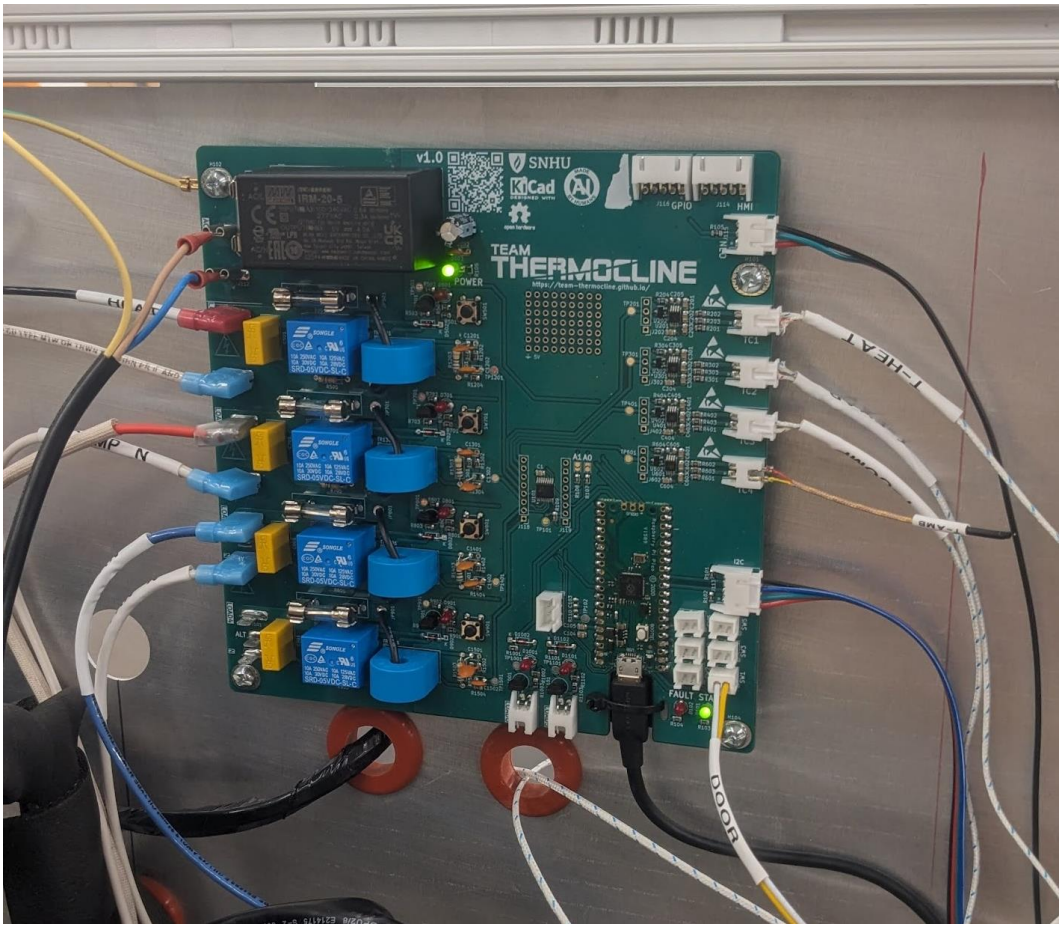

Figure 4; Main Controller PCB

Purpose and Design Description

The electrical subsystem provides power distribution, sensing, and interfacing for the system. It enables operation of the PCB during testing, supports temperature and humidity sensing, and provides connectivity for external data logging.

Requirements Satisfaction

This subsystem satisfies:

- COM-01 through maintaining PCB operation
- COM-02 and COM-03 through NEC-compliant grounding and wiring
- COM-04 through UL-compliant component selection

- COM-05 through properly rated sensors
- COM-06 through PC connectivity for data logging

Subsystem Performance

Only one PCB was ordered; we overbuilt enough parts of the board that we did not need to expand to a second revision. The source files were shared and reviewed by faculty, experts in the field, online communities, and fellow classmates. The main schematic is embedded into the documentation now but is also still viewable online.

4.4 Control and Software Subsystem

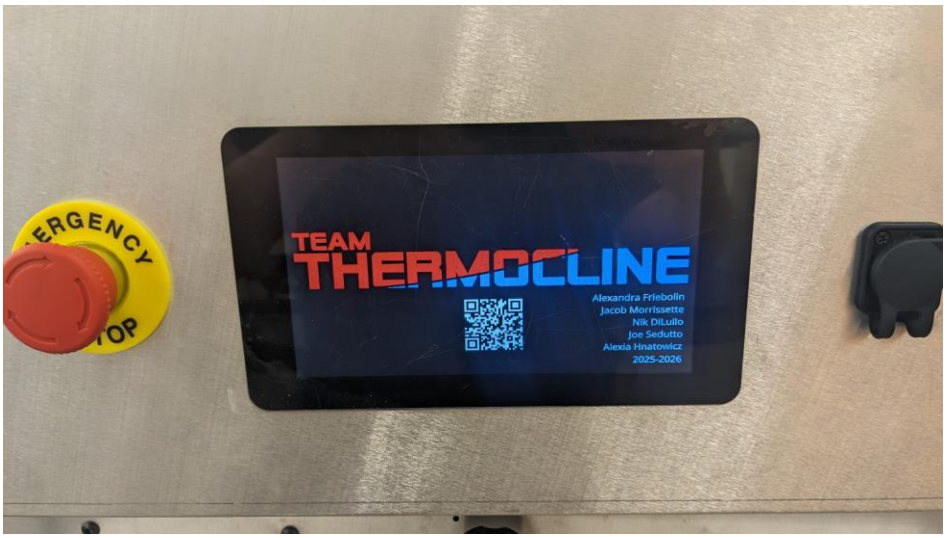

Figure 5; Front HMI

Purpose and Design Description

The control and software subsystem regulates temperature, manages system states, and provides user interaction. The system supports both local and remote temperature control and includes multiple operating modes such as shutdown, standby, and normal operation. Data logging for temperature, humidity, and system faults is integrated.

Requirements Satisfaction

This subsystem satisfies:

- CON-01 through CON-03 via temperature control and accuracy
- CON-04 and CON-05 via safety interlocks
- CON-06 through CON-08 via data logging
- CON-09 through CON-12 via system modes and control

Trade Study – HMI Selection

Option	Pros	Cons
External Computer	Easy to implement	Requires external device

Embedded LVGL Screen	Compact, integrated	Moderate programming complexity
Internal Computer System	Powerful UI	High complexity, cost

Table 4; Trade Study; HMI

Decision:

We experimented with an embedded crowpanel HMI but ultimately used a raspberry pi 5 as the HMI due to its ease of integration.

4.5 Safety Subsystem

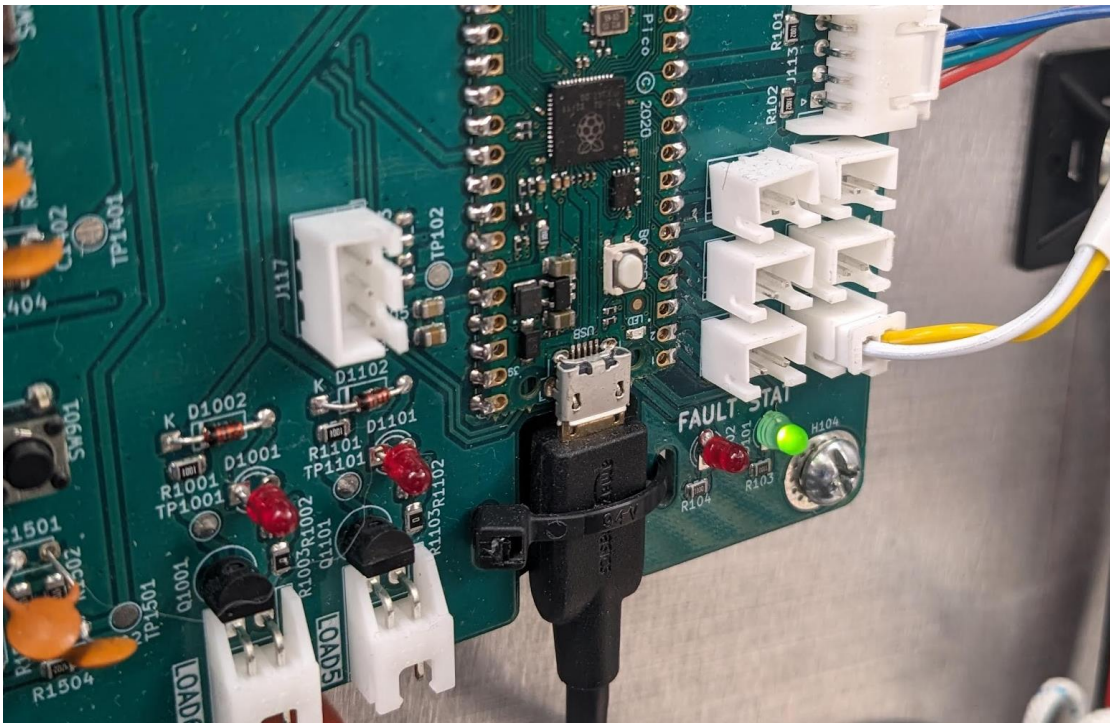

Figure 6; Controller PCB Safety Features

Purpose and Design Description

The safety subsystem ensures safe operation for both users and electronic components. It includes insulation, thermal protection, system interlocks, and defined safe operating states. External surfaces are designed to remain below safe-touch temperatures, and the system includes fault detection and automatic shutdown mechanisms.

Requirements Satisfaction

The safety subsystem satisfies:

- SAF-01 through material compatibility
- SAF-02 through external temperature limits
- SAF-03 through ambient operation capability
- SAF-04 through over-temperature protection
- SAF-05 through implementation of a safe state

Trade Study – Insulation Material Selection

Material	Thermal Conductivity	Cost	Malleability	Notes
Fiberglass	0.032–0.036 W/m·K	Low	High	Industry standard
Polyurethane Foam	Lower conductivity	Moderate	Moderate	Processing required
Polystyrene	Moderate	Low	Limited	Less flexible
Silica Aerogel	Very low	Extremely high	Low	Impractical

Table 5; Trade Study; Insulation Material

Final Decision (Important – align with your chosen design section):

While fiberglass demonstrated slightly better thermal performance, **denim insulation was ultimately selected** for implementation due to safety considerations during assembly and handling. Fiberglass poses risks such as skin irritation and airborne fibers, whereas denim insulation provides a safer, more user-friendly alternative with only a minimal reduction in thermal performance. This decision improves overall safety without significantly impacting system efficiency.

Safety in Software

Many safety systems were implemented in software, including those for current monitoring (link) as well as other safety sensors such as door switches, on and off timer interlocks and the E-Stop (link).

The safety firmware runs as a hardware watchdog, it must complete a cycle in a specific number of milliseconds otherwise a **hardware** function called a watchdog will force the CPU to reset itself. One of the first things the CPU does on reset is reconfigure all its IO low, thus, anytime the safety system is not running, for any reason, the fallback is to shut everything off and return to a true “safe mode”.

When the safety system is running however, the status light will flash in time with the heartbeat of the system. And this ensures to the user that the controller is performing all its safety tasks on time and many times per second. Software safety limits the maximum time loads can be on or off depending on needs. Ie, the heater cannot be on for more than 40 seconds, otherwise a fault is triggered. Indicating that a sensor may be disconnected if the PID truly thinks it needs to hold such a temperature for that long. The minimum off time of

the compressor is also set, the safety task ensures no hot-restarts will occur by monitoring and gating the compressor power. Restricting its restart time even if the PID desires it to be on instantly.

5 Fabrication and Integration

There were minor changes implemented in semester 2.

- Inner chamber rear-joint was changed from a seam joint to a butt join with riveted backplate. (link)
- Some gaskets originally not planned were added. (link)
- HMI Screen was swapped for the Raspberry Pi 5. (link)

6 Final System Test Results and Discussion

The final system did meet all requirements Including safe touch.

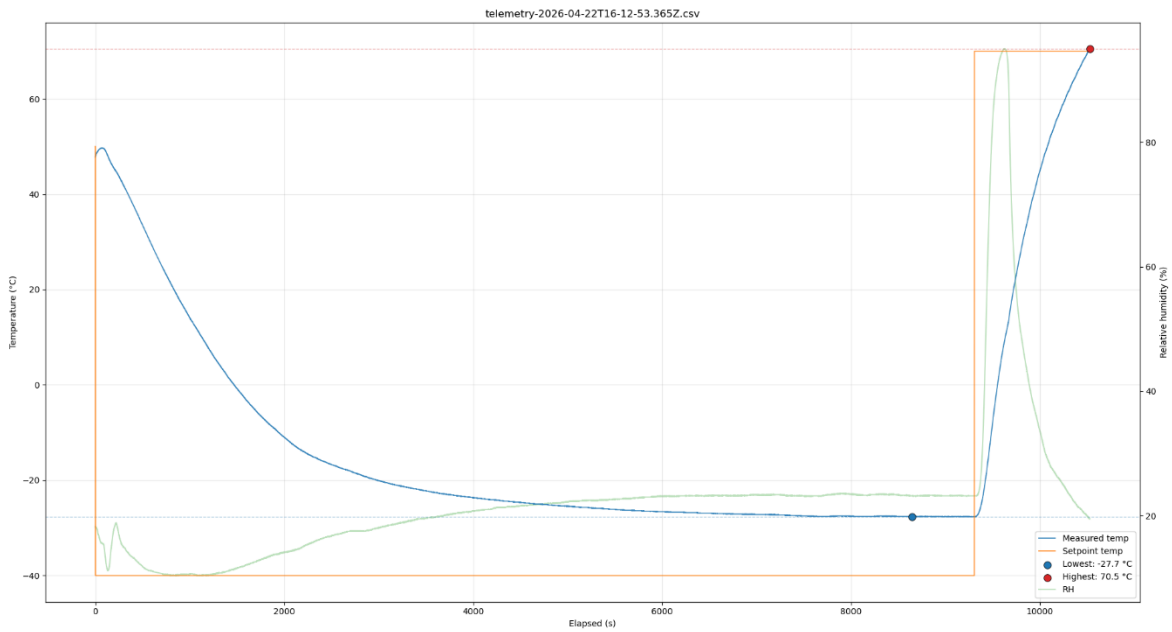

Figure 7; Heating & Cooling Data

Our primary requirements, to get the air temp inside a 12”x12” space to go from 0C to 70C was exceeded. The chamber was able to go from **–27C** to no less than **85C.** Surpassing both of our primary driving requirements.

We had designed the system to exceed the requirements by an even greater margin, constructed to withstand –40 to 85. Reasons we did not achieve true –40 were mainly due to poor gaskets and sealing at such high temperature differentials.

7 Lessons Learned

Gaskets: Gaskets and seals were consistently trickier to get working in practice. Small tests with low temperature differentials did not reveal that at higher extremes, our gaskets struggled to maintain good seals between door panels, body panels and the exterior. We tried no less than three different gasket materials and beleive futher research and experimentation with even more materials could yield better results.

Collaboration: Having a good definition of subsystems and distributed work helped to spread out the load. It was hard to have everyone collaborate on one aspect. But when broken up, multiple people could work on totally independent components. Creating well defined “interfaces” between parts of the system allowed for us to build off each other.

8 Recommendations

Control System: The control system as written is primarily focused on being gentle with the components. It purposefully leaves a lot of performance on the table. Safety restrictions are overbuilt, both for humans and for protecting the motors, heaters and other components. That's not to say that they're exactly as safe as they need to be. It is very likely they are overspecified and underloaded. Better tuning of the control system may be able to squeeze significant performance gains from the hardware by tuning limits that match the true capabilities of the hardware.

Sealing and Gaskets: Further experimentation and changing of gaskets may yield better sealing and thus, better results and lower temperatures.

9 Appendices

Appendix A: Schedule

The following table shows the team's main project milestones throughout both semesters.

Milestone	Type	Date Completed
Concept of Operations	Team	9-17-25
System Requirements Review (SRR)	Team	9-23-25
System Requirements Database	Team	9-23-25
Feasibility Test Plan	Team	10-21-25
Preliminary Design Review (PDR) Presentation	Team	11-18-25
Subsystem Reports	Individual	12-4-25
Critical Design Review (CDR) Report	Team	12-9-25
CDR Presentation	Team	12-11-25
Semester 1 Data Acceptance Package (DAP)	Team	12-18-25
Trade Studies	Team/Individual	12-18-25
Final Summary Presentation	Team	12-18-25
Delta CDR Presentation	Team	1-15-26
Verification Matrix & I&T Flow Diagram	Team	1-20-26
Subsystem Operational Description	Individual	2-10-26
Interim Report Presentation	Team	2-12-26
Preliminary Test Readiness Review (PTRR) Presentation	Team	2-26-26
System Functional Test Procedure & Demonstration	Team	3-17-26
Test Readiness Review (TRR) Presentation	Team	4-2-26
Final System Performance Test	Team	4-16-26
Poster Presentation	Team	4-23-26

Table 6; Aggregate Project Milestones

Appendix B: Financial Summary

The following table shows the team's final finances throughout the entire project's course of two semesters, from September 2025 to April 2026.

Item	Memo	Cost	Link	Date	Sybsystem
JLC order	Prototype PCB (HMI adapter)	$ 35.00		?	Electrical
SCS order	Prototype Send Cut Send Order (handle)	$ 30.00		?	Mechanical
HMI Screen	ESP32 based prototype HMI screen	$ 50.00		?	Electrical
Recovery Pump	Recovery pump tool	$ 388.00		?	Tools
Receiver Cylinder	Refrigerant containment cylinder	$ 49.00		?	Tools
Manifold Gauge	Refrigerant manifold gauge set	$ 66.00		?	Tools
Heater Core	Resistive Heater Core	$ 25.55		?	Electrical
9" fan	Prototype LCSC Order	$ 23.17		?	Electrical
Polycarb Sheet		$ 12.69		?	Mechanical
LCSC Order	LCSC order of electrical parts	$ 83.54		Friday, January 16, 2026	Electrical
JLC order	v1.0 pcb revision	$ 96.00	link	Friday, January 30, 2026	Electrical
McMaster Order 1	Mechnical miscellaneous (fasteners, frame pieces, etc.)	$ -		Tuesday, January 27, 2026	Mechanical
	i2c Humidity (SHT35)	$ 14.99	Grove Link	Friday, February 13, 2026	Electrical
	USB-C to USB 3.0	$ 14.99	Amazon	Friday, February 13, 2026	Electrical
	Screen 3 (7", touchscreen)	$ 65.95	Canakit	Friday, February 13, 2026	Electrical
	RPI4 2g	$ 66.79	Amazon	Friday, February 13, 2026	Electrical
	Copper Access Ports	$ 16.91	Amazon	Friday, February 13, 2026	Mechanical
	LED Light	$ 13.99	Amazon	Friday, February 13, 2026	Electrical
	E-Stop	$ 9.99	Amazon	Friday, February 13, 2026	Electrical
	2x20 10A	$ 5.99	Amazon	Friday, February 13, 2026	Electrical
	2x20 15A	$ 5.99	Amazon	Friday, February 13, 2026	Electrical
	Thermocouples	$ 13.99	Amazon	Friday, February 13, 2026	Electrical
	5 Cond. Wire (22 gauge)	$ 12.99	Amazon	Friday, February 13, 2026	Electrical
	IEC With Fuse and Switch	$ 7.99	Amazon	Friday, February 13, 2026	Electrical
Send Cut Send Order 1	6 parts, 2 duplicates inner chamber and front panel	$ 364.57	link	Thursday, February 26, 2026	Mechanical
Send Cut Send Order 2	All new parts yay	$ 438.41	Link	Monday, March 9, 2026	Mechanical
Send Cut Send Order 3	6 panels (Front Door Rear, Fan, Partition, 2x Side, Rear Vent)	$ 354.42	https://te	Thursday, March 12, 2026	Mechanical
McMaster Order 2	Mechanical miscellaneous (mounting hinges, pulling handles, leveling foot mounts, grommets, thumb screws)	$ 68.19		Thursday, March 12, 2026	Mechanical
Insulation	Denim insulation (safe touch)	$ 78.57		Thursday, March 12, 2026	Mechanical
McMaster Order 3	Purchase Order 3 (misc: aluminum tubing, capillary copper tubing, 10-32 rivet bolts/nuts, 1/4-20 bolts, resistive heater)	$ 150.25		Monday, March 23, 2026	Mechanical
Send Cut Send Order 4	4 panels (Top A, Top B, Bottom, Strap)	$ 310.31		Tuesday, March 24, 2026	Mechanical

Table 7; Aggregate Purchasing Orders

Total	% of budget	Left to spend		Metrics	
$ 2,874.23	95.81%	$ 125.77	**Breakdown**		
			Electrical	$	546.91
			Mechanical	$	1,824.32
Total (Not Including tools)			Tools	$	503.00
$ 2,371.23	79.04%				
			Supplier		
			JLCPCB	$	131.00
			LCSC	$	108.69
Mechanical	1824.32	175.68	SendCutSend	$	1,647.96
Electrical	546.91	-46.91	McMasterCarr	$	218.44
Tools	503		Other	$	849.56

Table 8; Total Budget Statistics